

Polycystic Ovary Syndrome — Clinical Reference

Endocrine disorder, 8–13% of reproductive-age people with ovaries · TeachIM.org (May 2022, optimized Jun 2026) · CC BY-NC 4.0 · educational use only — clinician-review before teaching

1 · Diagnosis — Rotterdam (2 of 3) + exclusion

Criterion	Definition
Hyperandrogenism	Clinical (hirsutism, acne, androgenic alopecia) OR biochemical (↑ androgens). Serum misses 20–40%
Ovulatory dysfunction	Oligomenorrhea (<8 cyc/yr or >35 d) or amenorrhea (>3 mo)
Polycystic ovaries	>12 follicles and/or volume >10 cc on US (or AMH in adults — 2023)

- **Diagnosis of EXCLUSION** — confirm only after ruling out mimics. Hirsutism = androgen-pattern terminal hair; acne/alopecia alone unreliable. Voice deepening/clitoromegaly → think tumor.

2 · Workup (exclude mimics)

Feature	Mimics	Tests
Hyperandrogenism	Non-classic CAH, acromegaly, androgen tumor, Cushing's	Total testosterone ; ±17-OHP; ±SHBG, DHEAS; midnight salivary cortisol if Cushingoid
Menstrual irregularity	Pregnancy, thyroid, prolactinoma, POI, structural	TSH±fT4, β-hCG, prolactin ; ±pelvic US; ±LH/FSH

- Total testosterone is the single best androgen test (free T unreliable). ↑ LH/FSH supports PCOS; its absence doesn't exclude it.

3 · Comorbidity screening (ALL patients)

Domain	Screen → manage
Insulin resistance / T2DM (7.5–10× risk)	A1c / fasting glucose / 2-h OGTT q1–2y → lifestyle ± metformin
Cardiovascular (↑LDL, ↓HDL)	BP + fasting lipids → statin per risk
Mood / eating (depression, anxiety, binge)	PHQ-9, GAD-7 → refer
Also	OSA, NAFLD per symptoms

4 · Gynecologic management

- **Endometrial protection (3× CA risk)**: withdrawal bleed ≥ every 90 days — OCP, progestin IUD, or cyclic progesterone. No routine US screening.
- **Hirsutism/cycles**: combined OCP (less-androgenic progestin: norgestimate, desogestrel, drospirenone) ± spironolactone 50 → 100 mg BID (teratogenic — contraception required).
- **Infertility** (most common anovulatory cause): 5–10% weight loss; letrozole/clomiphene for ovulation; refer OB-Gyn/REI early.
- **Pregnancy**: ↑ early loss, GDM (BMI-independent), pre-eclampsia, preterm — monitor closely.

5 · 2023 guideline update (framework unchanged)

- **AMH** may replace ultrasound for PCOM in ADULTS (not adolescents).
- Irregular cycles + hyperandrogenism (~70%) → no US/AMH needed.
- Adolescents: require BOTH hyperandrogenism + ovulatory dysfunction; no US/AMH.
- Anti-obesity pharmacotherapy incl. **GLP-1 RA** may be considered for weight (limited PCOS evidence).

Sources: Rotterdam/ESHRE-ASRM; Teede 2023 International Evidence-Based Guideline (Hum Reprod/JCEM/Fertil Steril); ACOG PB 194 (2018); Goodman AACE/ACE/AES 2015. Adapted from TeachIM.org under CC BY-NC 4.0.